

# Shashwat Agrawal

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## TECHNICAL SKILLS

**Languages:** C, C++, Rust, Python, JavaScript, SQL

**Frameworks:** Flask, Django, Node.js

**Developer Tools:** Git, Docker, CMake, Podman, NeoVim, Arch Linux

**Libraries:** Pandas, Matplotlib, Numpy, Tokio, Axum

## EXPERIENCE

### AI Tutor SWE

xAI

*Remote(Contractor)*

*May 2026 – Present*

- Evaluate and compare code outputs from multiple AI models across C, C++, Rust, and Python tasks, ranking responses based on code correctness, efficiency, and response quality.
- Conduct structured code reviews to assess model performance on systems programming tasks, identifying edge cases, bugs, and stylistic issues in generated code.
- Provide detailed, criteria-based rankings and feedback to inform model training and improvement at xAI.

### Open Source Contribution

*Pandas*

- Refactored `format.is_iso` by replacing an explicit loop-based ISO datetime check with a compiled regex, reducing Python iteration overhead and improving performance. [PR]
- Fixed incorrect ISO week 53 handling in `to_datetime`, preventing silent rollover for non-existent ISO weeks, ensuring standards-compliant behavior for ISO calendar years with only 52 weeks, and adding regression tests [PR]
- Aligned datetime parsing documentation with the PDEP-4 deprecation of `infer_datetime_format`, reflecting updated behavior and preventing user reliance on deprecated APIs. [PR]

## PROJECTS

### `tec.h` | C/C++

- Built a header-only, zero-dependency unit testing framework for C/C++ using macro meta-programming for automatic test discovery, registration, and rich assertions.
- Added test suites, fixtures (setup/teardown), expected failures (XFAIL), skipping, and advanced filtering (positive/negative, file-level).
- Implemented rich assertions (equality, floats, strings, comparisons), colored output, and C++ exception testing.
- Actively maintaining with recent additions like exclusion filters and CI improvements.

### Byte Machine | Rust

- Designed and implemented an 8-bit virtual machine with a custom ISA and 16-bit address space, complete with CPU emulation and memory management system.
- Engineered a custom two pass assembler that translates assembly code to bytecode, featuring support for label resolution, variables, direct memory addressing, and basic control flow.
- Implemented key components such as a stack, registers, and memory management, mimicking real-world processor functionality.
- Designed the VM and assembler for performance and extensibility, enabling easy addition of new opcodes and architectural features.

### HTTP Server | Rust, Tokio, Multi-threading

- Developed a high-performance **multi-threaded and asynchronous HTTP server** leveraging **Tokio** for concurrency.
- Achieved a **100% success rate** handling **100,000 requests** at 50 concurrent requests/sec.
- Optimized response times ranging from **400µs to 32.8ms**, with an average of **5.9ms** and peak throughput of **8,423 requests/sec**.

## EDUCATION

### Vishwakarma Institute of Information Technology (VIIT)

*B.Tech in Information Technology, CGPA: 8.36*

Pune, Maharashtra

*Sep 2024 – July 2027*

### SiddhiVinayak Technical Campus

*Diploma in Computer Science and Engineering, Percentage: 89.83%*

Khamgaon, Maharashtra

*Sep 2021 – July 2024*